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Realizing highly efficient vertical coupling with dielectric deflective metasurfaces

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Using dielectric deflective metasurfaces, we propose a novel, to the best of our knowledge, out-of-plane modulation scheme to realize vertical coupling on a 220 nm silicon-on-insulator platform. The metasurface is used to deflect vertical incident light to an oblique angle with high efficiency in the cladding layer. This deflection introduces a lateral wave vector component, thus preventing bi-directional transmission of traditional vertical coupling due to the second-order Bragg reflection of the grating. Additionally, an apodized design is employed for the subwavelength grating to improve mode matching with a deflection angle incident. The integration of the metasurface and subwavelength grating enables a new vertical coupling scheme with high efficiency. After global optimization, we achieved a simulation coupling efficiency of −2.19 dB. The measured coupling efficiency is −3.36 dB with a center wavelength of 1545.6 nm and a 1-dB bandwidth of 32 nm. The results confirm the feasibility of the proposed new architecture.

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Metasurfaces [1–3] are ultra-thin, two-dimensional structures composed of subwavelength optical components. They exhibit unprecedented engineering capability for manipulating the phase, amplitude, polarization, and orbital angular momentum of light. Over the past decades, metasurfaces have been extensively studied for their capabilities in realizing optical devices such as deflectors [4,5], metalens [6,7], holograms [8,9], and vortex wave generation [10]. These capabilities have found applications in optical display and sensing [11], hyperspectral imaging [12], and polarization camera [13]. In recent years, people have combined silicon photonics with metasurfaces to design new devices. By fabricating metasurfaces on top of the waveguide layer, guided waves can be molded into any desired free-space modes to achieve complex free-space functions [14]. Based on this platform, guided wave-driven metasurfaces have been experimentally demonstrated with novel effects, such as out-of-plane focusing with polarization multiplexing [15], on-chip triple-channel holographic multiplexing [16], and emission and manipulation of higher-order Poincaré sphere beams [17].

As a typical silicon photonic device, vertical grating couplers (VGCs) function as optical I/O interfaces that convert vertically incident light into in-plane light [18–20]. They offer the advantage of not requiring angled polishing of coupling fibers [21], making them advantageous in wafer-level testing [22] and multicore fiber packaging [23]. Recently, metasurfaces have also provided new impetus for the optimization of a grating coupler. People have implemented on-chip metalens to reduce the taper length from about 100 to 30 μm [24] and even 5 μm [25]. Nevertheless, as an important property of VGCs, the coupling efficiency (CE) is limited by the second-order Bragg reflection effect, which can only achieve approximately 34% with a binary grating structure on a 220 nm silicon-on-insulator (SOI) platform [26]. To further enhance CE, researchers have introduced special L-shape grating [27], anti-reflect structure [28] or shift-pattern structure [29] to compensate the diffraction effect, which however, requires a two-step etching process with a high alignment accuracy in the sub-hundred-nanometer range. This coupling architecture will reduce the devices' robustness in fabrication and limits their further applications in optical I/O systems.

In this Letter, we propose a novel out-of-plane modulation scheme to realize vertical coupling, which combines a dielectric metasurface with an apodized binary grating to achieve high-efficient vertical coupling on the 220 nm SOI platform. By introducing phase discontinuities at the interface, dielectric metasurface deflects vertically incident light to the target angle with near unity transmission in the C-band. Then an apodized single-etched grating coupler (SEGC), which incorporates subwavelength grating (SWG) structures to enhance mode matching, is optimized to collect the deflected oblique light with high efficiency. Notably, the integration of the metasurface and SEGC can easily compensate the second-order Bragg reflection effect. The single-etched grating is designed with a minimum feature size of 140 nm, which is compatible with most commercial silicon photonics fabs. The coupling efficiency is −3.36 dB (−2.19 dB) in the experiment (simulation) for TE-polarized vertical incidence at a wavelength of 1546.5 nm with a 1-dB bandwidth (BW) of 32 nm. Furthermore, the 1-dB angular tolerance and 1-dB displacement of the device are also measured as approximately 3° and 6 μm in this study.

As elaborated in Fig. 1, the overall device consists of a deflected silicon metasurface and an apodized single-etched grating coupler. The process of receiving vertically incident light into the waveguide device can be understood in two steps. Initially, TE-polarized light enters from the air and interacts with the nanopillars comprising the metasurface, resulting in deflection through the SiO_2 cladding (Fig. 1(a)). Subsequently, the obliquely incident light is captured by the SEGC and efficiently

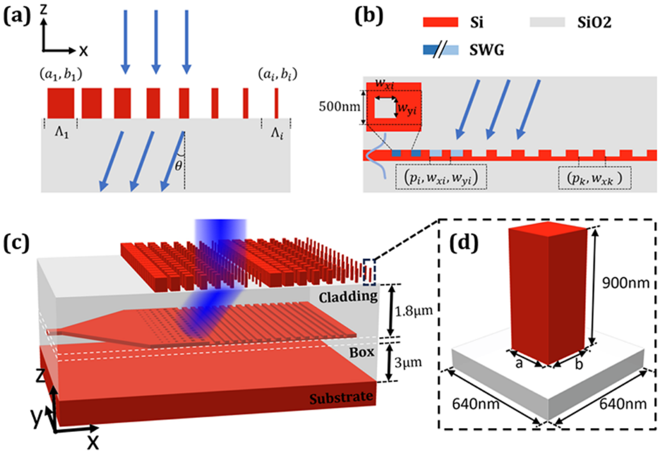

Fig. 1. (a) Operation of a deflective metasurface. (b) Single-etched grating coupler. (c) Vertical grating coupler integrated with a deflective metasurface. (d) Meta unit.

coupled into the waveguide (Fig. 1(b)). The overall structure of the device is shown in Fig. 1(c). By introducing a lateral wave vector component, the deflection prevents the vertical coupling from the bi-directional transmission. This new scheme enables the straightforward realization of vertical coupling designs based on a binary grating structure.

For the metasurface design, we employed a square periodic boundary with each side measuring 640 nm, and α-Si nanopillar with a height of 900 nm as the fundamental unit, as illustrated in Fig. 1(d). The parameters "a" and "b" denote the width and length of nanopillars, respectively. The structures were simulated in FDTD Solutions, a commercial software package from Lumerical Inc. In the case of the TE-polarized incident light from air, we simulated the transmission phase and efficiency for different parameters, as depicted in Figs. 2(a) and 2(b). According to the results, we design a deflective metasurface, and the deflection angle introduced can be expressed as [1]

$$\sin(\theta_t)n_t - \sin(\theta_i)n_i = \frac{\lambda_0}{2\pi}\frac{d\Phi}{dx},$$

where λ_0 is the wavelength of incident and n_t is the refractive index of the cladding layer. θ_i and n_i stand for the air's refractive index (1) and the incidence angle (0°), respectively. Φ denotes the transmission phase of metasurface elements. In our design, 9 different units are chosen, with a minimum linewidth constraint of 140 nm, to distribute the 2π phase gradient, resulting in a period length Λ of 5.76 µm and a deflection angle θ_t in the cladding layer of 10.83°. The angle, which is about 16° in air, will be utilized in the grating optimization, mimicked by an oblique Gaussian mode source. The distribution of E-field of the metasurface and far field intensity in the cladding layer is plotted in Figs. 2(c) and 2(d), indicating a deflection efficiency of 90%, with a transmission of 97%. The parameters of the designed metasurface deflector are shown in Table 1.

The grating design was implemented on a 220 nm SOI platform, with a 3 µm-thick buried oxide layer. The SEGC consists of 6 SWG regions and 21 ridged waveguide regions with an etch depth of 100 nm. As shown in Fig. 1(b), the SWG is a rectangular etching array with a period of 500 nm in the y-direction. The optimization of mode matching of the SWG region with the source can be achieved by adjusting the groove width in both directions

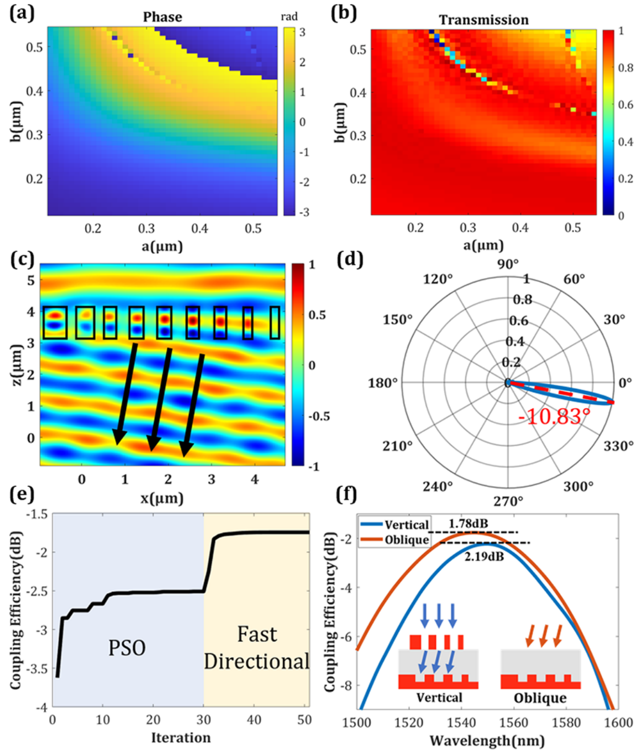

Fig. 2. (a) Phase and (b) transmission map for unit cells. (c) Normalized E-field distribution and (d) farfield intensity of a deflective metasurface. (e) Iteration result of the two-stage optimization. (f) Simulation of GC with oblique incident and VGC with vertical incident.

Table 1. Parameters of the Metasurface[a]

a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8	a_9
170	170	180	150	280	220	230	390	520
b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8	b_9
260	320	350	440	430	420	470	410	430

[a]Unit:nm.

(w_{xi}, w_{yi}) of the etched rectangle and the grating period (p_i). The SWG region can be equivalent as the effective medium described by the duty cycle [18,30], which allows the full-wave simulation in 2-dimensional FDTD solver. A Gaussian source with a diameter of 10.4 µm was employed to simulate the outcoming light of the single mode fiber (SMF). For the SEGC design, an initial uniform grating is obtained using particle swarm optimization (PSO) for an oblique Gaussian source, assuming w_{yi} is 250 nm. The fast directional optimization algorithm [31] is then employed to rigorously optimize the parameters of each grating cycle, with a minimum linewidth constraint of 180 nm. The iteration process of the whole optimization is shown in Fig. 2(e). After approximately 50 iteration cycles, the optimal parameters for the apodized grating design were achieved, resulting in an insertion loss of 1.8 dB at a wavelength of 1550 nm. The simulation results in Fig. 2(f), show that the CE of the SEGC can reach −1.78 dB at 1545 nm with oblique incidence. Finally, through the integration of the deflected metasurface and the apodized grating, the proposed vertical coupling device achieves a simulated efficiency of 2.19 dB at 1550 nm with vertical incidence, demonstrating the feasibility of the new vertical

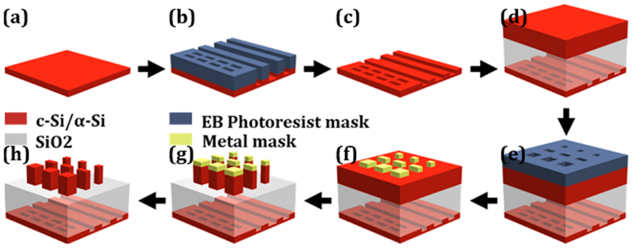

Fig. 3. Fabrication process of VGC. (a)–(c) Fabrication process of apodized single-etched grating. (d) Deposition of SiO_2 and α-Si layer. (e)–(h) Fabrication process of deflective metasurface.

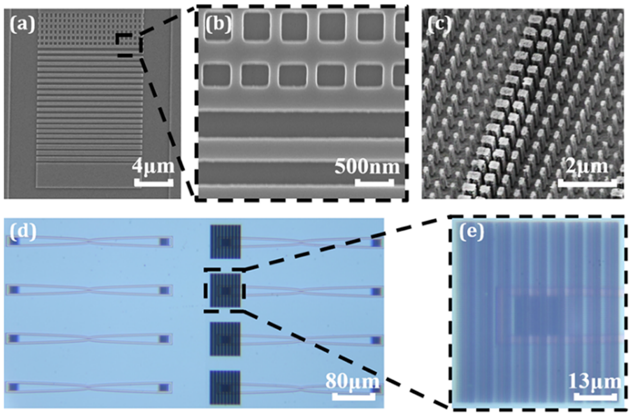

Fig. 4. (a) SEM image of single-etched GC. (b) Detailed view of the SWG area. (c) SEM image of metasurface nanopillars. (d) Mircoscopy view of VGC. (e) Detailed view of the VGC covered with metasurface.

coupling scheme. The extra insertion loss around 0.41 dB is mainly induced by the deflection and transmission loss of metasurface. Overall, the simulation results above demonstrate the feasibility of the new vertical coupling scheme.

The manufacturing process of the proposed VGC is illustrated in Fig. 3. Figures 3(a)–(d) show the production process of the GC. The grating is fabricated on a 220 nm SOI wafer. After a standard Electron Beam Lithography (EBL) process a 100 nm shallow Si etch is performed. The 1.8 μm SiO_2 cladding and 900 nm α-Si film were grown by the Plasma-Enhanced Chemical Vapor Deposition (PECVD) technology sequentially. We fixed the parabolic taper length to be 110 μm. A grating width of 12 μm was chosen for mode matching with the SMF. Figures 3(e)–3(h) demonstrate the production process of the deflective metasurface. The metasurface is fabricated on the α-Si layer with EBL and etching. The sample size is adjusted to a size of 50 μm × 50 μm in order to fully cover the grating area. When the alignment error occurs, the metasurface corresponding to the grating region has a consistent phase gradient, resulting in an alignment-insensitive response. Figure 4(a) shows SEM images of the grating coupler, while Fig. 4(b) provides more details of the SWG region. The fabricated metasurface is presented in Fig. 4(c), from which we can clearly observe Si nanocolumns with nine different sizes. Figures 4(d) and 4(e) show the optical images of whole VGC samples.

For the measurement setup, a SANTEC TSL-550 tunable laser is utilized as the light source, and its polarization is restricted to TE mode through adjustment of the polarization controller. For

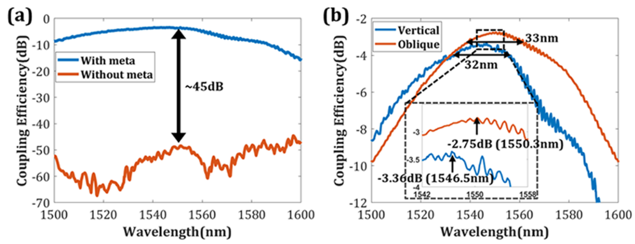

Fig. 5. (a) Experimental results of GCs with and without meta under vertical incidence. (b) Experimental results of our VGC under vertical incidence and the reference GC under oblique incident.

Table 2. Comparison of Figure of Merits of Binary VGCs

Ref.	CE (dB) Sim.	CE (dB) Exp.	1-dB BW (nm)	Polarization	Grating Height (nm)
[23]	−2.0	−2.7	33	TE	380
[26]	−3.77	−4.69	~30	TE	220
[19]	−3.0	−5.8	-	TE	220
[20]	−6.20	−6.58	-	TM	340
This work	−2.19	−3.36	32	TE	220

the VGC sample test, the incident light is vertically coupled into the sample with the metasurface. The collected light was then measured by a YOKOGAWA AQ2200-221 optical power meter. A wavelength scan was performed from 1500 to 1600 nm with a 0.2 nm step. The grating samples with and without the metasurfaces were both evaluated under vertical incident. Compared to the reference sample without a metasurface, Fig. 5(a) shows a 45 dB significant enhancement in the CE when the metasurface is integrated. A more detailed test result in Fig. 5(b) shows that the VGC has a peak CE of −3.36 dB at 1549 nm with a 1-dB bandwidth of about 32 nm (blue solid line) under vertical incidence. The reference grating under oblique incidence has a peak CE of −2.75 dB at 1550 nm with a 1-dB bandwidth of about 33 nm (orange solid line). Compared to their simulation results, the test results exhibit an extra loss of approximately 1 dB, mainly owing to the fabrication error of the SEGC.

A comparative analysis of recent studies on vertically coupled binary gratings is presented in Table 2, highlighting the performance metrics. It is evident that the integration of metasurfaces allowed us to achieve a high coupling efficiency (−3.36 dB) on a 220 nm SOI platform under vertical incidence. This novel coupling scheme provides a new methodology of out-of-plane modulation to overcome the challenges associated with second-order Bragg reflection in the traditional vertical grating coupler design.

To further characterize the performance of the device, the tolerance of the incident angle and x axis displacement is tested. The simulated variation trend of the coupling curve of the incidence angle from −2° to 4° is depicted in Fig. 6(a). It can be observed that with an increasing angle, there is a decrease in the central wavelength of the coupling curve from 1565 to 1515 nm. As shown in Fig. 6(b), the experimental results align well with the simulation findings, demonstrating a range of 1-dB angle tolerance between −1° and 2°. The impact of the displacement in the x axis on the coupling efficiency curve is depicted in Fig. 6(c), which shows a 1-dB displacement that corresponds to approximately 6 μm along the x axis. Experimental results

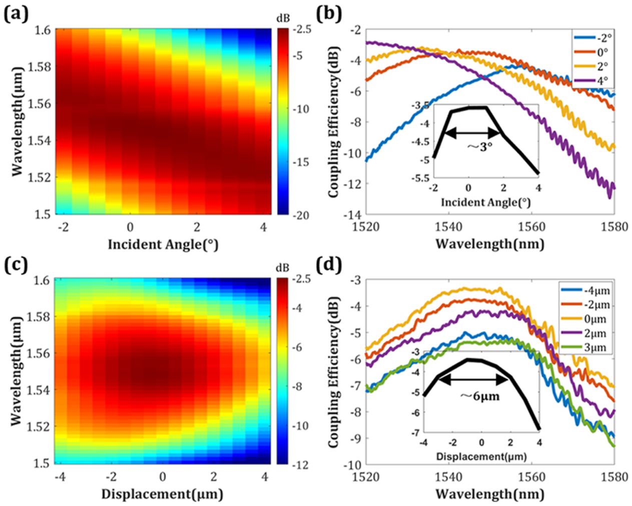

Fig. 6. (a) Simulated and (b) measured coupling efficiency with different incident angles from −2° to 4°. The inset plot in (b) depicts the relationship between the CE and the angle of incident at the central wavelength. (c) Simulated and (d) measured efficiency with different displacements from −4 to 4μm. The inset figure in (d) illustrates the correlation between the CE and *x* axis displacement at the wavelength of 1550 nm.

demonstrate that the fabricated sample exhibits a 1-dB displacement of about 6 μm, aligning with the curves ranging from −4 to 2 μm as depicted in Fig. 6(d).

In conclusion, a new vertical grating coupling scheme combined with deflective metasurfaces is experimentally demonstrated on the 220 nm SOI platform. Using a dielectric metasurface, vertically incident light can be deflected to the target angle, which can be collected by the apodized single-etched grating coupler with high efficiency. The measured coupling efficiency is 3.36 dB at 1546.5 nm with a 1-dB bandwidth of 32 nm. Further characterization tests reveal the architecture's angular tolerance, with approximately 3° tolerance for angular deviations, and *x* axis displacement tolerance of 6 μm. Subsequent research works will go deeper into the phase distribution and freedom of degrees of the metasurface, which may develop additional functions and channels into the vertical coupler device, such as polarization splitting grating coupler and wavelength-division-multiplex receiver.

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Data availability. The data underlying the results presented in this Letter are not publicly available at this time but may be obtained from the authors upon reasonable request.

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